

1 · Development Method

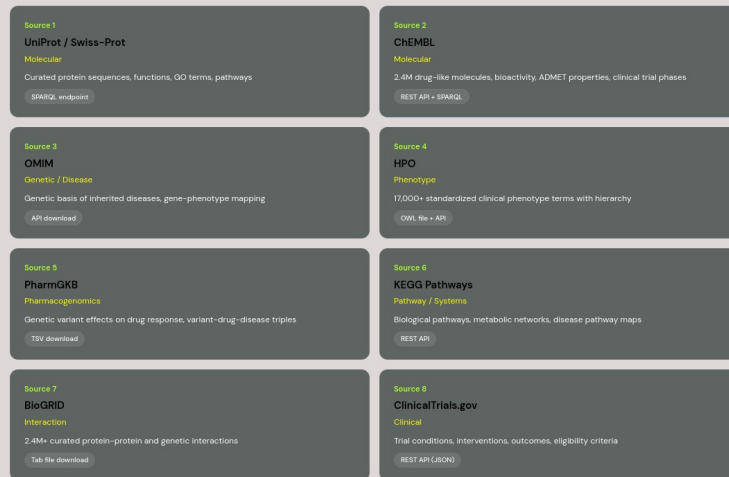
LLM-Augmented Semantic Knowledge Graph (LASKG) — a 5-stage pipeline alternative to PrimeKG's manual ETL



2 · Data Resources

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8 open-access biomedical repositories — SPARQL-enabled, multi-scale coverage



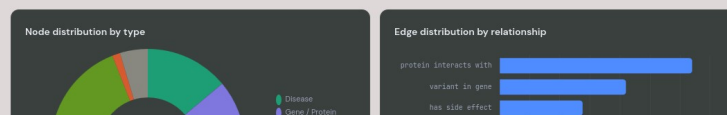
3 · Node Types

8 biological entity categories spanning molecular to clinical scales



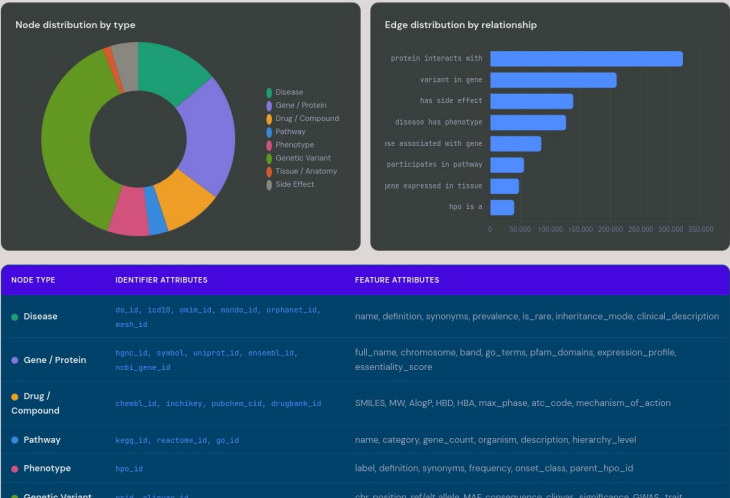
4 · Attributes

Feature-rich node and edge properties for ML readiness



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Feature-rich node and edge properties for ML readiness



5 · Relationships

18 directed edge types — 1173,400 total relationships

#	SOURCE	RELATIONSHIP	TARGET	EVIDENCE	EDGES	VOLUME
11	Gene	protein_interacts_with	Gene/Protein	BioGRID / STRING	320,000	
8	Variant	variant_in_gene	Gene/Protein	Ensembl VEP	210,000	
9	Drug	has_side_effect	Side Effect	SIDER / FAERS	138,000	
7	Disease	disease_has_phenotype	Phenotype	HPO annotations	126,000	
1	Disease	disease_associated_with_gene	Gene/Protein	OMIM / GWAS catalog	85,000	
6	Gene	gene_participates_in_pathway	Pathway	KEGG / Reactome	56,200	
13	Gene	gene_expressed_in_tissue	Tissue	GTEx / HPA	48,000	
15	Phenotype	hpo_is_a	Phenotype	HPO hierarchy	40,000	
18	Drug	drug_drug_interaction	Drug	DrugBank / TWOSIDES	30,000	
17	Disease	comorbid_with	Disease	EHR / CPRD	25,000	
2	Drug	is_indication_for	Disease	CHEMBL / ClinicalTrials.gov	22,400	
9	Variant	variant_causes_disease	Disease	ClinVar / GWAS	18,000	
4	Drug	drug_targets_gene	Gene/Protein	CHEMBL / DGIdb	16,600	
14	Pathway	pathway_dysregulated_in	Disease	DisGeNET / KEGG	12,000	
12	Disease	disease_affects_anatomy	Tissue	UBERON / DisGeNET	9,400	
16	Drug	drug_modulates_pathway	Pathway	LLM + PubMed	7,800	
10	Variant	variant_affects_drug_response	Drug	PharmGKB	6,200	
3	Drug	is_contraindicated_in	Disease	CHEMBL / FDA labels	4,900	

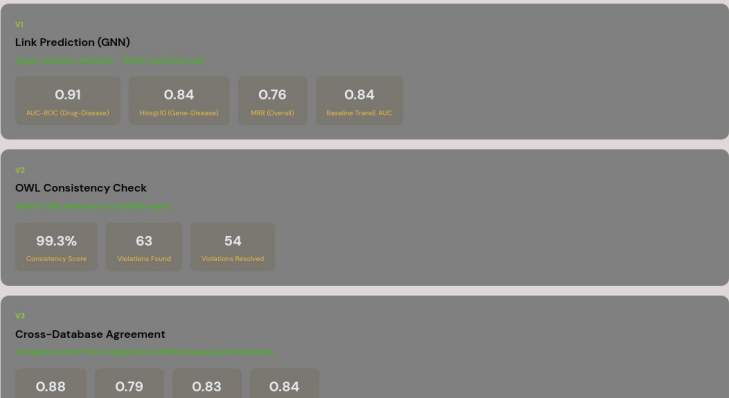
6 · Interactive Knowledge Graph

Force-directed network — drag nodes, hover for details, filter by type, switch hub views



7 · Validation Results

5 complementary validation strategies ensuring graph quality and biological relevance



BioKG-Alt vs PrimeKG

Side-by-side comparison showing key differences in approach

DIMENSION	PRIMEKG (ORIGINAL)	BIOKG-ALT (OURS)
Construction	Manual ETL + Python scripts	Ontology-first + SPARQL + LLM
Data Sources	20 fixed databases	8 open-access SPARQL repositories
Node Types	10 major biological scales	8 typed node categories
Total Edges	4,090,249	1,173,400 (high-confidence)
Literature Mining	Not included	LLM/NLP from PubMed
Update Mechanism	Full rebuild	Continuous CI/CD pipeline
Semantic Reasoning	Not available	OWL reasoner (HermiT)
Edge Provenance	Source database only	Source + confidence + PMID + method
Validation	Cross-database overlap	5 strategies: GNN + OWL + expert + CCB + downstream
Storage	CSV flat files	Neo4j + RDF/Turtle dual format
Pharmacogenomics	Limited	Full PharmGKB integration
Clinical Trials	Not included	ClinicalTrials.gov API integrated

